



The switches (S1 through S8) are used by players and are extended through wires. The LEDs (LED1 through LED8) are fixed on the front side of the boxes. These LEDs can be covered and labeled with the names of the players written on them for identification. When the power is switched on by connecting to 230V AC, the quiz buzzer circuit is ready to use.

platform for eight players/teams to play at a time.

Additionally, the piezo buzzer (PZ1) can also produce a sound when transistor T2 conducts. At the same time, transistor T1 conducts to ground pin 11 of IC2, latching all the Q0 through Q7 outputs. This restricts further changes in the output state due to any changes in the state of switches S1 through S8 by any other players. Only one of the eight LEDs glows and buzzer sounds until the circuit is turned off and then back on.

An actual-size, single-side PCB layout for the quiz competition buzzer circuit is shown in Fig. 3, and its component layout is in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box. Fix all nine LEDs (LED1 through LED9) on the front side of the box, and the piezo buzzer on the rear side. Label LED9, which indicates that the power is on, meaning the circuit is enabled. Assign LED1 through LED8 to players/teams.

Use twin flexible wires to extend the switches to the players' table and fix the switches there, allowing players to press the switches during the quiz petitions.

After properly installing the switches at the players' tables, connect them to the PCB of the circuit you assembled. Now, the quiz competition buzzer is ready to use.

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-quiz-competition-buzzer-for-eight-team> **EFY**

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